



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

SEPTEMBER 2007
YEARLY EXAMINATION
YEAR 11 Continuers

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.

Total marks—100 Marks

- Attempt all questions.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections: A, B, C.

Examiner: Mr C.Kourtesis

Section A — Start a new booklet

Marks

Question 1 (31 marks)

- (a) If $L = gt^2$, find the value of L if $g = 10$, $t = 3$. 1
- (b) Solve $2a + 1 = -7$. 1
- (c) Factorize
- (i) $3a + 15$; 1
- (ii) $x^2 - A^2$. 1
- (d) Simplify
- (i) $3(2a - 4)$; 1
- (ii) $\frac{n}{4} + \frac{n}{3}$; 1
- (iii) $10a^2b \times 2b^3$. 1
- (e) Solve
- (i) $|x - 4| = 14$; 2
- (ii) $-3x \geq 12$; 1
- (iii) $4x^2 = 5$. 1
- (f) Express $\frac{1}{\sqrt{10}}$ with a rational denominator. 1
- (g) If $f(x) = \frac{4}{x}$, find $f\left(\frac{1}{a}\right)$. 1
- (h) If $y = 7x^4 + x - 1$, find $\frac{dy}{dx}$. 2
- (i) Find the exact value of: $\tan 150^\circ + \cos 45^\circ$. 2

(j) The line $6x - ky = 4$ passes through the point $(2, -1)$. Find the value of k .

2

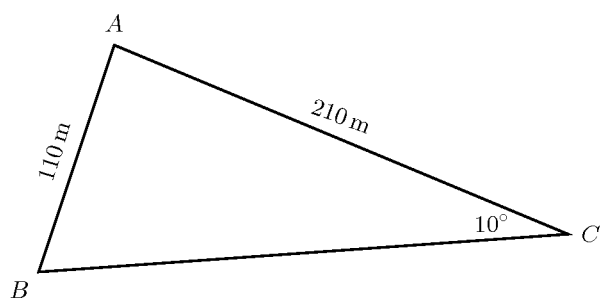
(k) Solve the pair of simultaneous equations

2

$$y = 2x + 1$$

$$x - 2y = 7$$

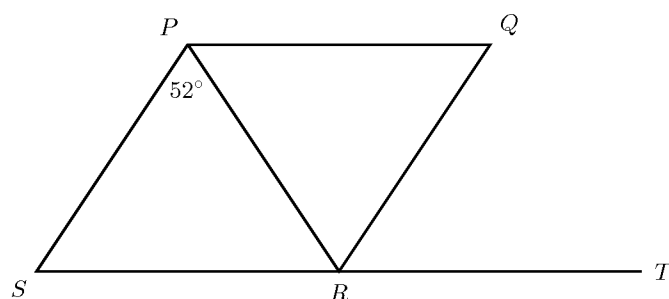
(l)



2

Find the size of $\widehat{ABC}$ to the nearest degree.

(m)



$PQRS$ is a rhombus where $\widehat{SPR} = 52^\circ$ and SR is produced to T .

(i) Find the size of $\widehat{SPQ}$.

1

(ii) What is the size of $\widehat{QRT}$? Give reasons.

2

(n) If $\sqrt{45} + \sqrt{80} = a\sqrt{5}$, find the value of a .

2

(o) Find

(i) $\log_2 8$;

1

(ii) $\log_3 54 - \log_3 18$.

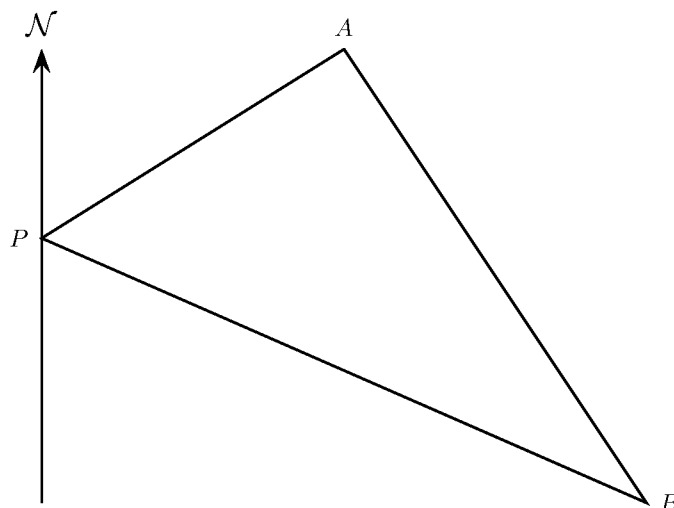
2

Section B — Start a new booklet

Question 2 (35 marks)

- (a) Express $0.\dot{2}\dot{4}$ as a rational number. 1
- (b) Factorize $x^3 - 27$. 1
- (c) Solve the equation 2
- $$\tan \theta = -\frac{1}{\sqrt{3}} \text{ for } 0^\circ \leq \theta \leq 360^\circ.$$
- (d) The first three terms of an arithmetic series are 12, 17, and 22. Find:
- (i) the thirty-fifth term of the series; 2
- (ii) the sum of the first thirty-five terms. 1
- (e) A geometric series has second term 6 and the ratio of the seventh term to the sixth term is 3.
- (i) Find the common ratio, r . 1
- (ii) What is the first term, a ? 1
- (iii) Calculate the sum of the first 12 terms. 1
- (f) Differentiate with respect to x :
- (i) $(1 - 10x)^{11}$; 1
- (ii) $(2x + 1)(3 - x^2)$ using the product rule; 2
- (iii) $\frac{x}{x^2 + 4}$ using the quotient rule. 2
- (g) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$. Find the values of:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\alpha^2 + \beta^2$ 2
- (h) A parabola P has equation $x^2 = 8(4 - y)$. Find:
- (i) the coördinates of its focus; 2
- (ii) the equation of the directrix; 1
- (iii) the coördinates of all points of intersection of P with the coördinate axes. 1

(i)



Ship A is 35 nautical miles from a port P and is on a bearing of 065° . Ship B is 50 nautical miles from P and is on a bearing of 115° .

Find:

(i) the size of $\widehat{APB}$;

1

(ii) the distance between the ships (answer to the nearest nautical mile).

2

(j) A function $f(x)$ is defined by

$$f(x) = \begin{cases} 0 & \text{if } x \leq -2 \\ -1 & \text{if } -2 < x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

(i) Find $f(-4) + f(-\frac{1}{2}) + f(0)$.

2

(ii) Sketch the graph of $y = f(x)$.

1

(k) In a certain school the senior students are divided into two classes A and B . Class A contains 11 female and 9 male students and class B contains 8 female and 12 male students. A certain occasion requires two speakers, one from each class. These are selected at random. Find the probabilities that:

(i) both speakers are male;

1

(ii) both speakers are female;

1

(iii) one speaker is male and one is female.

2

(l) Find the domain and range of

2

$$y = \frac{1}{\sqrt{9 - x^2}}.$$

Section C — Start a new booklet

Question 3 (34 marks)

(a) Solve

(i) $|4 - 5x| > 1$

2

(ii) $|x + 1| = 2x + 7$

2

(b) Solve the equation

$$x^6 - 28x^3 + 27 = 0.$$

3

(c) Find the perpendicular distance from the point $(4, -3)$ to the line $5x = 2y + 9$.

2

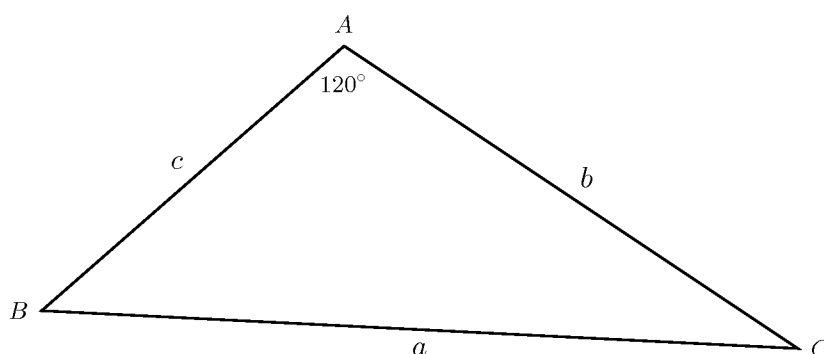
(d) In the Cartesian plane, indicate by shading the region whose points simultaneously satisfy $y \geq 1$ and $y \leq \sqrt{4 - x^2}$.

3

(e) If $n^2 \equiv an(n - 1) + bn + c$, find the values of a , b , and c .

3

(f)



3

In triangle ABC , prove that

$$\cos C = \frac{2b + c}{2a}.$$

(g) Two marksmen, A and B , fire simultaneously at a target. If A is twice as likely to hit the target as B , and if the probability that the target does get hit is $\frac{1}{2}$, find the probability of A hitting the target.

3

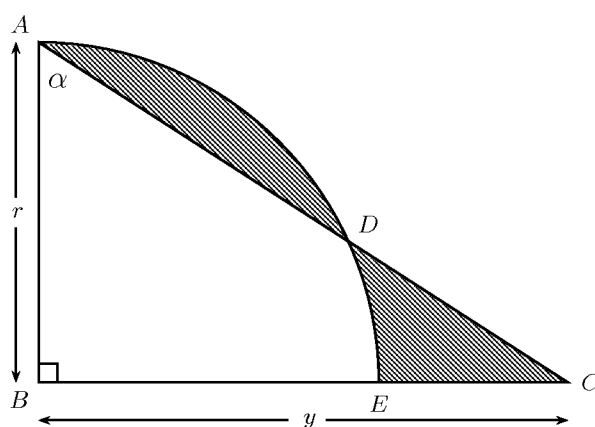
(h) If $\cot \theta = -\frac{5}{12}$ and $\cos \theta > 0$, find $\sin \theta$. 3

(i) The n^{th} term of the series $3 + \frac{4}{3} + \frac{11}{18} + \dots$ is $A \left(\frac{1}{3}\right)^n + B \left(\frac{1}{2}\right)^n$. Find
(i) the values of A and B ; 3

(ii) the fourth term. 1

(j) Find the equation of the parabola with axis $y = 0$, vertex $(0, 0)$, and the line $x - y + 3 = 0$ as a tangent to the curve. 3

(k) 3



$ADEB$ is a sector with centre B . $\triangle ABC$ is right-angled at B .
The two shaded regions have the same area.
Show that $\tan \alpha = \frac{\pi}{2}$.

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Question 1:

$$\begin{aligned} \text{a) } L &= 10 \times 3^2 \\ &= 10 \times 9 \\ &= 90 \end{aligned}$$

$$\text{f) } \frac{\sqrt{10}}{10}$$

$$\begin{aligned} \text{b) } 2a &= -8 \\ a &= -4 \end{aligned}$$

$$\begin{aligned} \text{g) } f(1/k) &= 4/\sqrt{1}a \\ &= 4a \end{aligned}$$

$$\text{c) i) } 3(a+5)$$

$$\text{h) } \frac{dy}{dx} = 28x^3 + 1$$

$$\text{ii) } (x+A)(x-A)$$

$$\text{i) } \frac{-1}{\sqrt{3}} + \frac{1}{\sqrt{2}}$$

$$\text{d) i) } 6a - 12$$

$$\text{ii) } \frac{3n+4n}{12} = \frac{7n}{12}$$

$$\begin{aligned} \text{j) } 12+k &= 4 \\ k &= -8 \end{aligned}$$

$$\text{iii) } 20a^2b^4$$

$$\begin{aligned} \text{k) } y &= 2x+1 & \textcircled{1} \\ x-2y &= 7 & \textcircled{2} \end{aligned}$$

$$\begin{aligned} \text{e) i) } x-4 &= 14 \\ x &= 18 \end{aligned}$$

① into ②

$$x-2(2x+1)=7$$

$$-x+4=14$$

$$x-4x-2=7$$

$$-x=10$$

$$-3x=9$$

$$x=-10$$

$$\boxed{x = -3}$$

sub into ①

$$\text{so } x = 18, -10$$

$$y = 2x-3+1$$

$$\boxed{y = -5}$$

$$\text{ii) } -3x \geq 12$$

$$x \leq -4$$

check in ②

$$\begin{aligned} -3-2x-5 &= -3+10 \\ &= 7 \checkmark \end{aligned}$$

$$\text{iii) } x^2 = 5/4$$

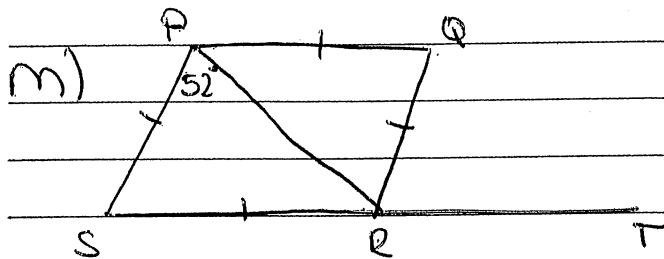
$$x = \pm \frac{\sqrt{5}}{2}$$

$$1) \frac{\sin \angle ABC}{210} = \frac{\sin 10}{110}$$

$$\sin \angle ABC = \frac{210 \sin 10}{110}$$

$$\angle ABC = 19.36046155$$

$$= 19^\circ \text{ (nearest degree)}$$



$PS = PQ = QR = RS$ (sides of a rhombus)

$\therefore \triangle PRS$ is isosceles

$\therefore \angle PRS = 52^\circ$ (base $\angle$'s isos $\triangle$)

$\therefore \angle PSR = 76^\circ$ ($\angle$ sum $\triangle$)

$\therefore \angle PQR = 76^\circ$ (opp $\angle$'s in rhombus)

$\therefore \angle QPR \text{ and } \angle QRP = 52^\circ$ (base $\angle$'s isos $\triangle$)
($\triangle SPR \equiv \triangle QPR$ SAS test)

$\therefore \angle SPQ = 104^\circ$

ii $\angle SRQ = 104^\circ$ (Opp $\angle$'s of Rhombus)

$\therefore \angle QRT = 76^\circ$ ($\angle$ sum st line)

$$n) \sqrt{9} \times \sqrt{5} + \sqrt{16} \times \sqrt{5}$$

$$= 3\sqrt{5} + 4\sqrt{5}$$

$$= 7\sqrt{5}$$

$$\therefore a = 7$$

$$o) i) \log_2 8 = 3$$

$$ii) \log_3 54 - \log_3 18$$

$$= \log_3 \frac{54}{18}$$

$$= \log_3 3 = 1$$

Section B

2 (a) $0.2\dot{4} = \frac{24}{99} = \frac{8}{33}$ ①

(b) $(x-3)(x^2+3x+9)$ ①

(c) $\tan \theta = -\frac{1}{\sqrt{3}}$ quad 2 $\frac{180-\theta}{4}$ $\frac{180+\theta}{360-\theta}$

now $\tan 30^\circ = \frac{1}{\sqrt{3}}$

So answers are $180+30=210^\circ$ ①
 $360-30=330^\circ$ ①

(d) arithmetic 12, 17, 22

$a=12$

$d=5$

(i) $T_{35} = a + (n-1)d$
 $= 12 + 34 \times 5$
 $= 182$ ②

(ii) $S_{35} = \frac{n}{2}(a+L)$
 $= \frac{35}{2}(12+182)$ ①
 $= 3395$

(e) geometric $T_2 = ar = 6$

$\frac{T_7}{T_6} = \frac{ar^6}{ar^5} = 3$ $r=3$

(i) $r=3$ ①

(ii) $3 \times a = 6$ $a=2$ ①

(iii) $S_n = \frac{a(r^n-1)}{r-1} = \frac{2(3^{12}-1)}{3-1} = 531440$ ①

(f) (i) $\frac{d}{dx}(1-10x)^{11}$
 $= 11(1-10x)^{10} \times -10$
 $= -110(1-10x)^{10}$ ①

(ii) $\frac{d}{dx}(2x+1)(3-x^2)$
 $= (2x+1) \times -2x +$
 $(3-x^2) \times 2$
 $= -4x^2 - 2x + 6 - 2x^2$
 $= -6x^2 - 2x + 6$ ②

(iii) $\frac{d}{dx}\left(\frac{x}{x^2+4}\right)$
 $= \frac{(x^2+4) \times 1 - x \times 2x}{(x^2+4)^2}$
 $= \frac{x^2+4-2x^2}{(x^2+4)^2}$
 $= \frac{4-x^2}{(4+x^2)^2}$ ②

$$(9) x^2 - 5x + 2 = 0$$

$$a=1 \quad b=-5 \quad c=2$$

$$(i) \alpha + \beta = -\frac{b}{a} = \frac{5}{1} = 5 \quad (1)$$

$$(ii) \alpha\beta = \frac{c}{a} = \frac{2}{1} = 2 \quad (1)$$

$$(iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 25 - 4 = 21 \quad (2)$$

$$(h) x^2 = 8(4-y)$$

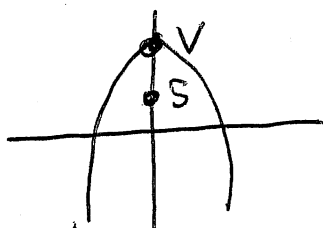
$$(x-0)^2 = -8(y-4)$$

$$(x-0)^2 = -4 \times 2(y-4)$$

$$(x-h)^2 = -4a(y-k)$$

(i) vertex is $(0, 4)$ $a=2$, down curve.

focus $(0, 2)$ (2)



(ii) directrix is $y = 6$ (1)

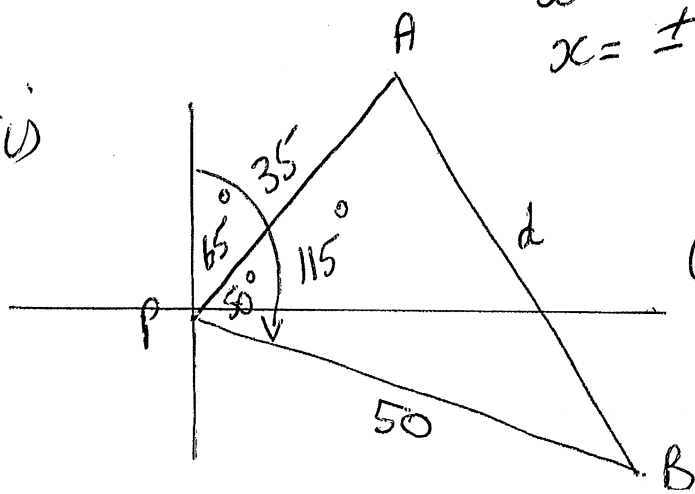
cuts y axis at $(0, 4)$

cuts x axis when $x^2 = 8(4-0)$

$$x^2 = 32$$

$$x = \pm\sqrt{32} = \pm 4\sqrt{2} \quad (1)$$

(i)



$$(i) \hat{APB} = 50^\circ \quad (1)$$

$$(ii) d^2 = 35^2 + 50^2 - 2 \times 35 \times 50 \times \cos 50^\circ$$

$$d \approx 38.41$$

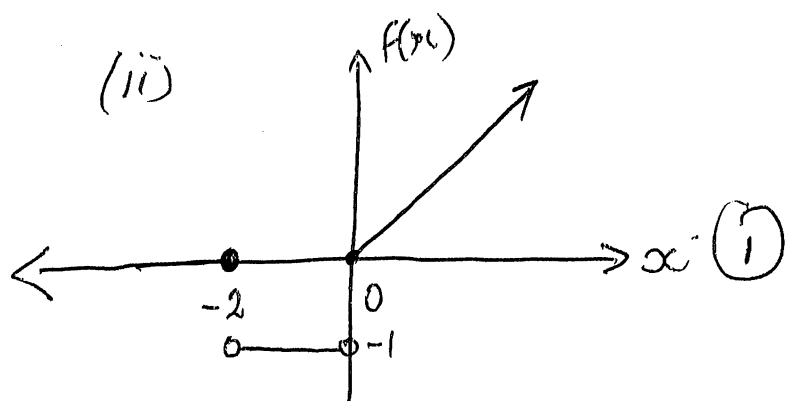
$$= 38 \text{ nm} \quad (2)$$

$$(1) \quad (i) \quad f(-4) = 0$$

$$f(-\frac{1}{2}) = -1$$

$$f(0) = 0$$

$$-1 \quad (2)$$



(k)

$\frac{A}{11F}$	$\frac{B}{8F}$
$9m$	$12m$

$$(i) P(mm) = \frac{9}{20} \times \frac{12}{20} = \frac{27}{100} \quad (1)$$

$$(ii) P(FF) = \frac{11}{20} \times \frac{8}{20} = \frac{11}{50} \quad (1)$$

$$(iii) \frac{9}{20} \times \frac{8}{20} + \frac{11}{20} \times \frac{12}{20} = \frac{61}{100} \quad (2)$$

$$(L) \quad y = \frac{1}{\sqrt{9-x^2}}$$

Domain $-3 < x < 3 \quad (1)$

Range if $x=0$, $y = \frac{1}{3}$ Range $y \geq \frac{1}{3} \quad (1)$

(3)

(a).

$$|4 - 5x| > 1$$

$$(i) \quad 4 - 5x > 1 \quad 4 - 5x < -1$$

$$\frac{5x < 3}{x < \frac{3}{5}} \quad \frac{5x > 5}{x > 1}$$

$$(ii) \quad |x+1| = 2x+7$$

$$\begin{aligned} x+1 &= 2x+7 & x+1 &= -2x-7 \\ x &= 6 & 3x &= -8 \\ & & x &= -8/3 \end{aligned}$$

$$(b) \quad x^6 - 28x^3 + 27 = 0$$

$$y^2 - 28y + 27 = 0$$

$$(y-27)(y-1) = 0$$

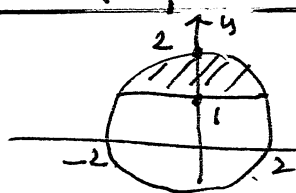
$$\therefore x^3 = 27, \quad x = 3$$

$$x^3 = 1, \quad x = 1$$

$$(c) \quad D = \frac{|5 \times 4 - 2 \times (-3) - 9|}{\sqrt{29}}$$

$$= \frac{17}{\sqrt{29}} \div 3.16$$

(d)

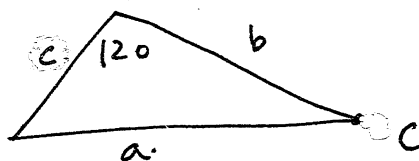


$$(e) \quad x^2 = ax^2 + (b-a)x + c$$

$$\therefore a = 1, \quad b-1 = 0 \therefore b = 1$$

$$c = 0$$

(f)



$$-\frac{1}{2} = \frac{b^2 + c^2 - a^2}{2bc}$$

$$a^2 = b^2 + c^2 + bc$$

$$\therefore c = \frac{a^2 + b^2 - c^2}{2ab}$$

Subst. ① into ②

$$\therefore c = \frac{2b + c}{2a}$$

$$(g) \quad P(A)P(B) + P(A)P(\bar{B}) + P(\bar{A})P(B) = \frac{1}{2}$$

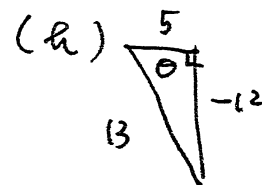
$$P(A) + P(B) = P, \quad P(B) = \frac{p}{2}$$

$$\frac{p^2}{2} + P(1 - \frac{p}{2}) + \frac{p}{2}(1 - P) = \frac{1}{2}$$

$$\therefore p^2 + 2p - p^2 + p - p^2 = 1$$

$$p^2 - 3p + 1 = 0$$

$$p = \frac{3 - \sqrt{5}}{2}, \quad \therefore p \leq 1$$



$$\sin \theta = \frac{-12}{13}$$

$$-0.923$$

$$(i) \quad 3 + \frac{4}{3} + \frac{11}{18} + \dots$$

$$= A\left(\frac{1}{3}\right)^n + B\left(\frac{1}{2}\right)^n$$

$$n=1, \quad \frac{A}{3} + \frac{B}{2} = 3$$

$$n=2, \quad \frac{A}{9} + \frac{B}{4} = \frac{4}{3}$$

Solve ① & ② $A=3, B=f$

$$(ii) \quad T_4 = 3\left(\frac{1}{3}\right)^4 + 4\left(\frac{1}{2}\right)^4 = \frac{31}{108}$$

$$(j) \quad y^2 = 4ax, \quad x = y-3$$

$$y^2 - 4ay + 12 = 0$$

$$\Delta = 0, \quad 16a^2 - 48 = 0$$

$$a = \sqrt{3}$$

$$\therefore y^2 = 4\sqrt{3}x$$

(k)



$$A(I) = A(II)$$

$$y = r \tan \alpha$$

$$I + III =$$

$$II + III = \frac{r^2}{2} \tan \alpha$$

$$I - II = 0$$